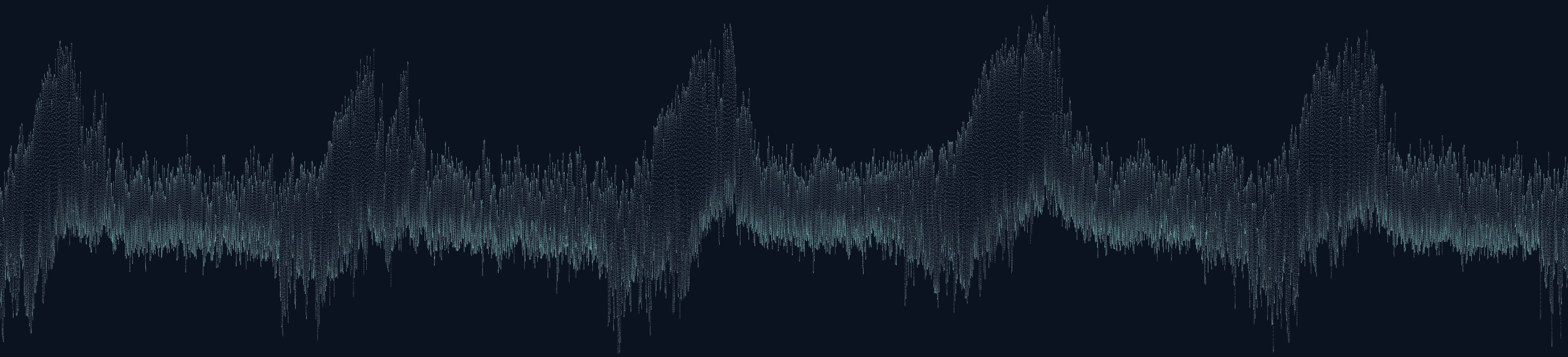
CE436 · DISCRETE-TIME SIGNAL PROCESSING · FINAL DEFENSE

Weather forecasting on FPGA

Parallel fixed-point inference and verification on a Cyclone II DE2 board



Every line below is real data: 43,824 hourly temperature observations, Ho Chi Minh City, 2021–2025 (NASA POWER)

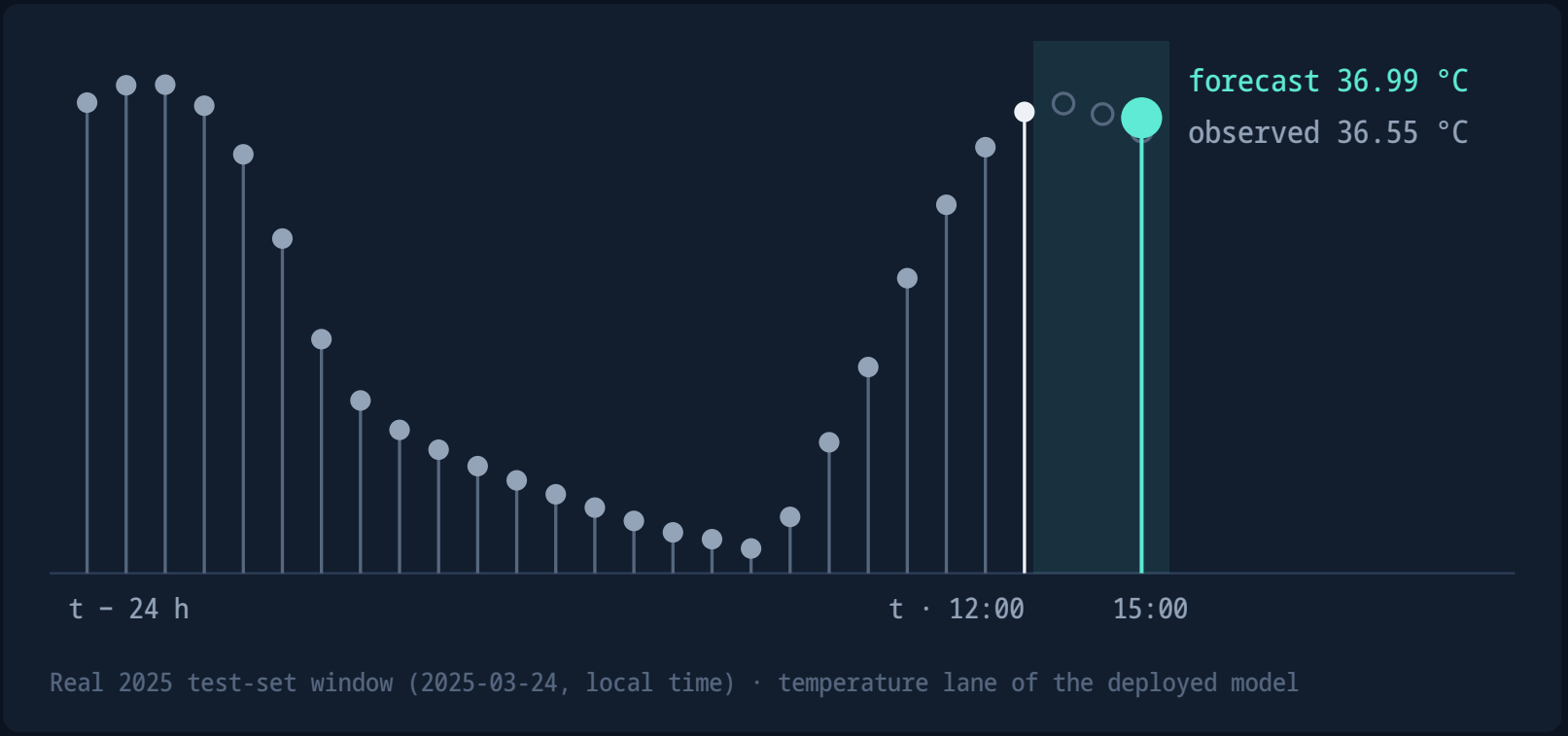
Forecast the weather three hours ahead

+3

HOURS

Five quantities from each hourly frame

- Temperature
- Pressure
- Humidity
- Wind speed
- Wind direction



Real 2025 test-set window (2025-03-24, local time) · temperature lane of the deployed model



From one temperature output to five weather quantities

1

forecast quantity

Temperature prototype

MODEL	Three-parameter seasonal OLS
ARITHMETIC	One lane · 16-bit Q8
DIRECTION	No direction path

5

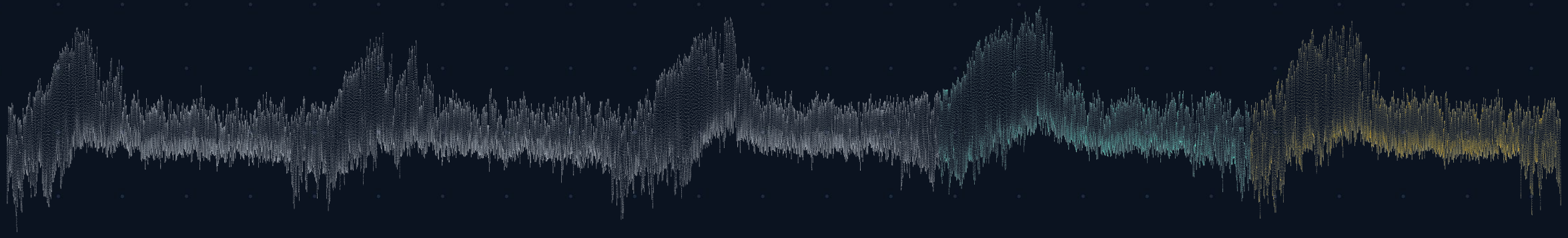
weather quantities

Current weather design

MODEL	Seven-lag regression per channel
ARITHMETIC	Six lanes · 18-bit Q8
DIRECTION	Cosine + sine, CORDIC angle

Unchanged: hourly inputs and the three-hour target horizon

A chronological split: the model never sees the future



2021–2023

26,253

Training

Fits the coefficients

2024

8,781

Validation

Selects the model

2025

8,757

Test

Held out, frozen model

Usable examples: each needs a valid 25-hour history and a target three hours later. NASA POWER hourly, Ho Chi Minh City, UTC.

Seven lags per channel, one linear sum

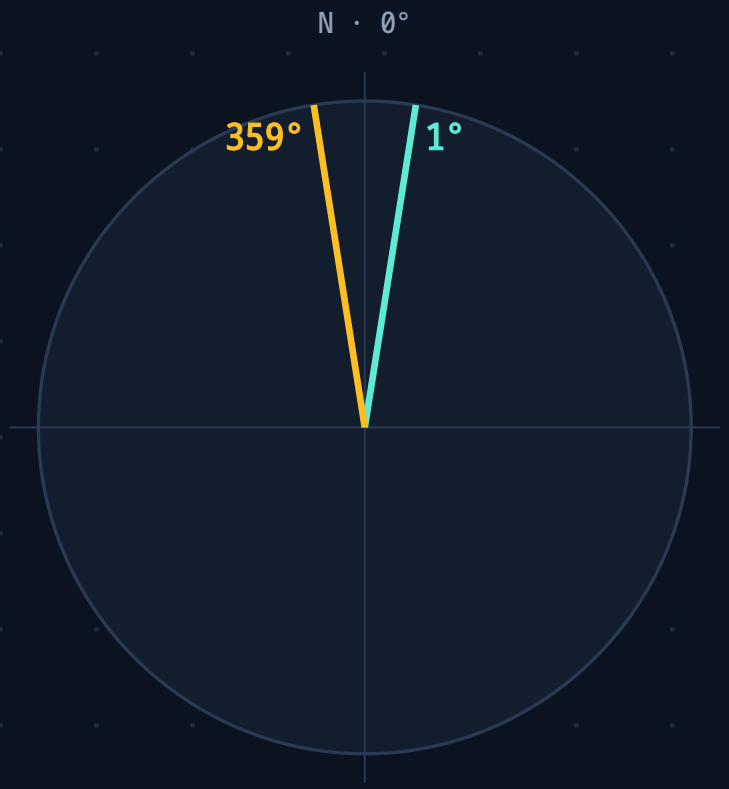
$$\hat{x}(t+3) = b + \sum w_k \cdot x(t - \ell_k)$$

k = 1 … 7 · each channel uses only its own history



Deployed temperature weights: the last hours and the same hours yesterday dominate

Wind direction wraps around at 0°



2° apart, not 358°

Rays drawn wider apart than 2° for legibility

ENCODE ON HOST → PREDICT ON FPGA → RECONSTRUCT

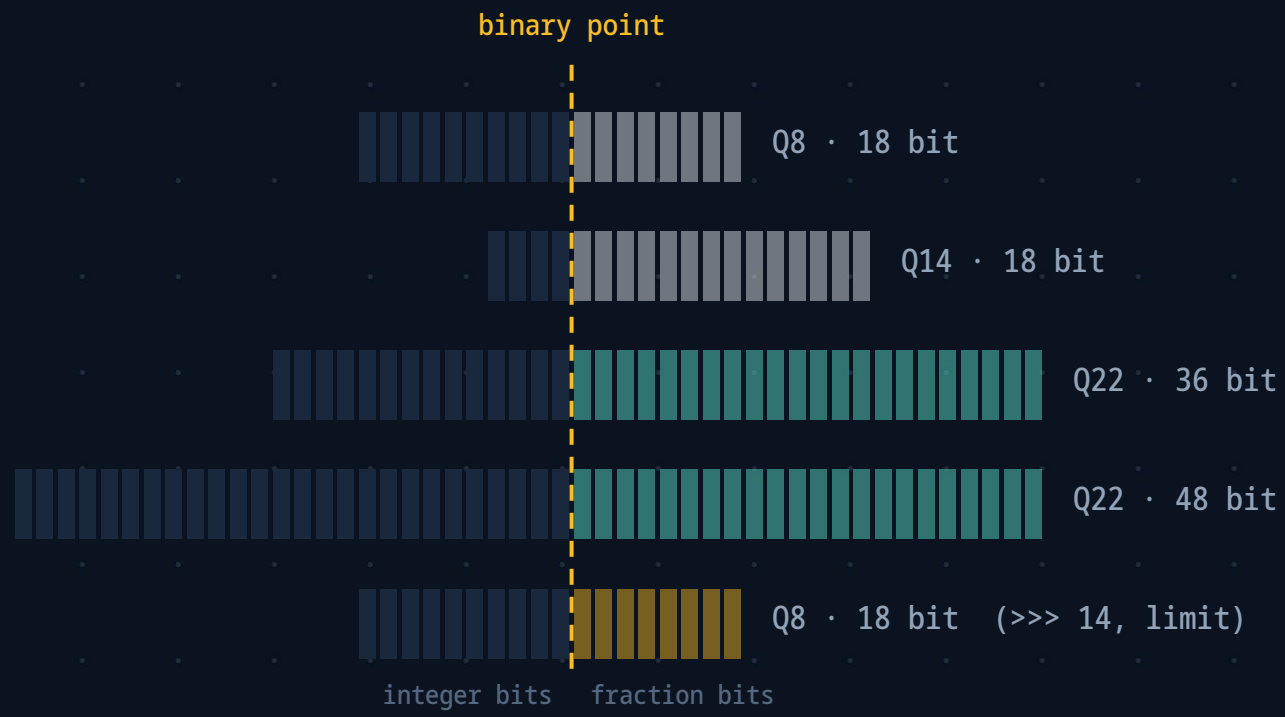
cos θ → lane 5 →
sin θ → lane 6 →

CORDIC
16 iterations → θ̂

Each angle becomes two smooth components, so nearby directions stay numerically close. The FPGA predicts both and turns them back into one angle.

`Zero component pair → direction flagged undefined`

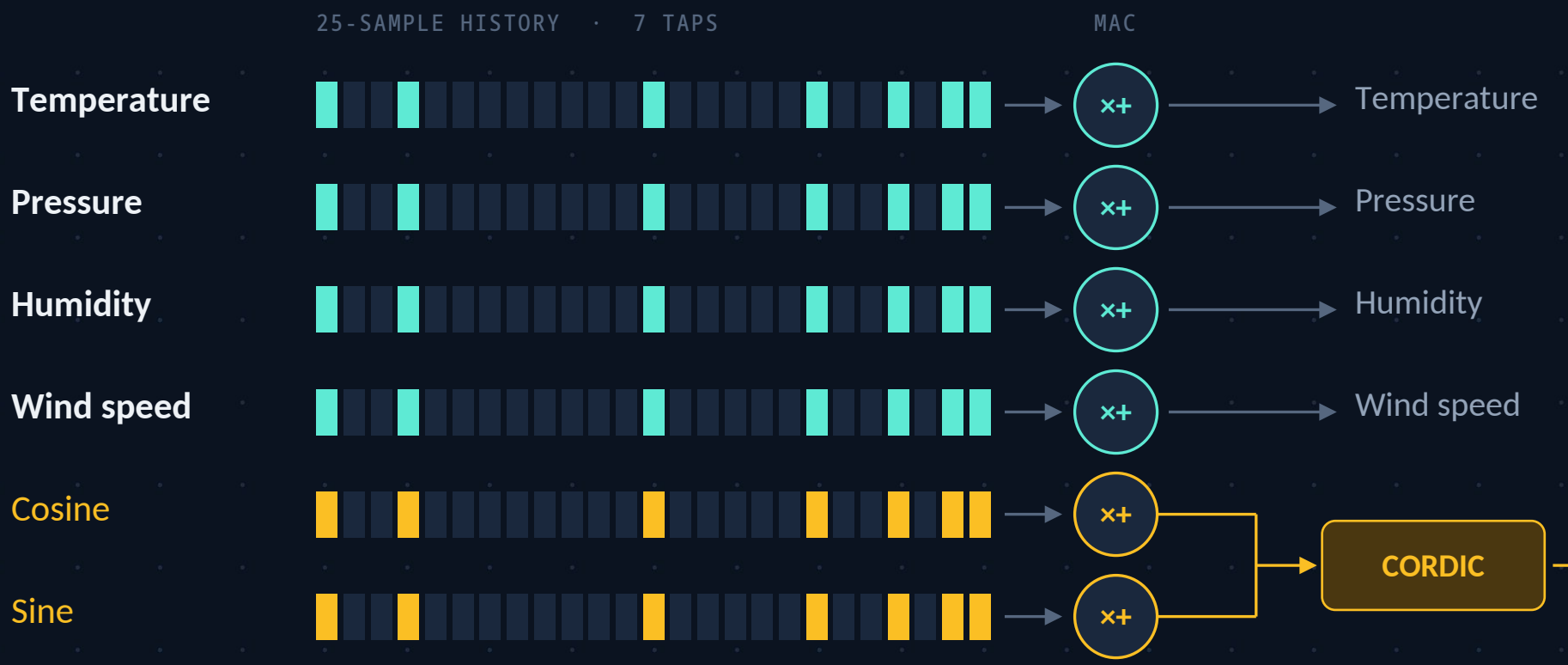
Fixed point: keep every bit until the final shift



$30\ °C \times 2^{8} = 7680\ \text{raw}$

Aligned bias, floor scaling, range limits. The C++ reference uses identical rules, so any mismatch is an implementation bug, not forecast error.

Six lanes in parallel, one multiplier each



6 **multipliers, not 42**

Each multiplier walks its seven taps over time.

History advances only when a frame is accepted. The wrapper aligns scalar results with the reconstructed direction.

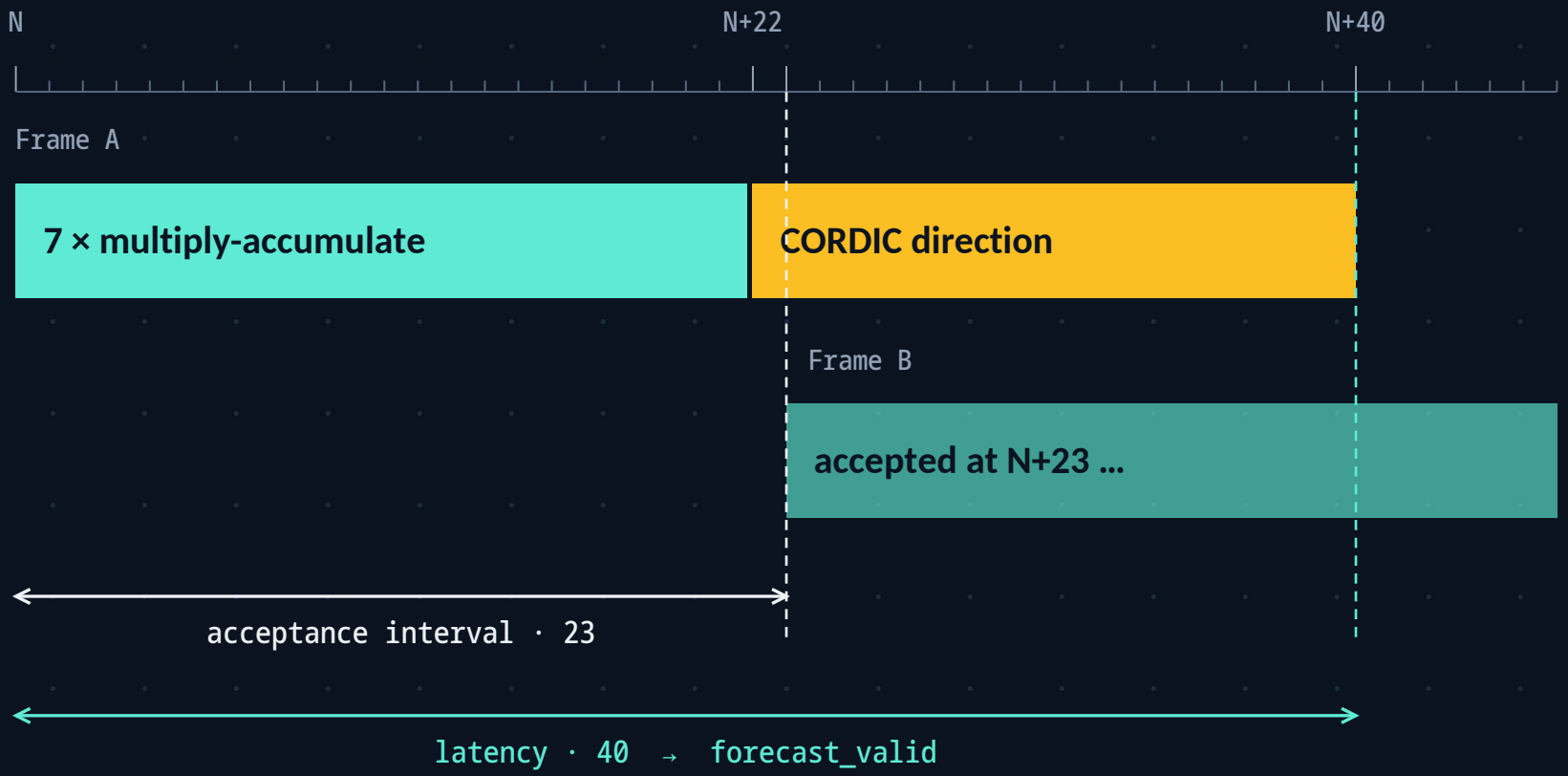
40 cycles of latency, a new frame every 23

40

CYCLES LATENCY

0.80 µs at 50 MHz

First forecast follows the 25th accepted frame: 24 frames fill the history.



An idle clock is not another hour: history shifts only on accepted frames.

Three independent checks, each closer to silicon

STAGE 1

RTL + DPI-C

1,124

forecasts checked

from 1,200 accepted frames

C++ golden model, cycle-level scoreboard

STAGE 2

Core SDF

264

forecasts checked

from 288 input frames

Gate-level netlist, max delays, 20 ns clock

STAGE 3

Physical JTAG

104

forecasts checked

from 128 input frames

Real responses read from the DE2 board

CHECKED AT EVERY STAGE

Values | Transaction order | Reset behavior | Cycle timing

The model beats both baselines on every quantity

Temperature
MAE · °C

−39%

vs best baseline

Pressure
MAE · kPa

−60%

vs best baseline

Humidity
MAE · pp

−31%

vs best baseline

Wind speed
MAE · m/s

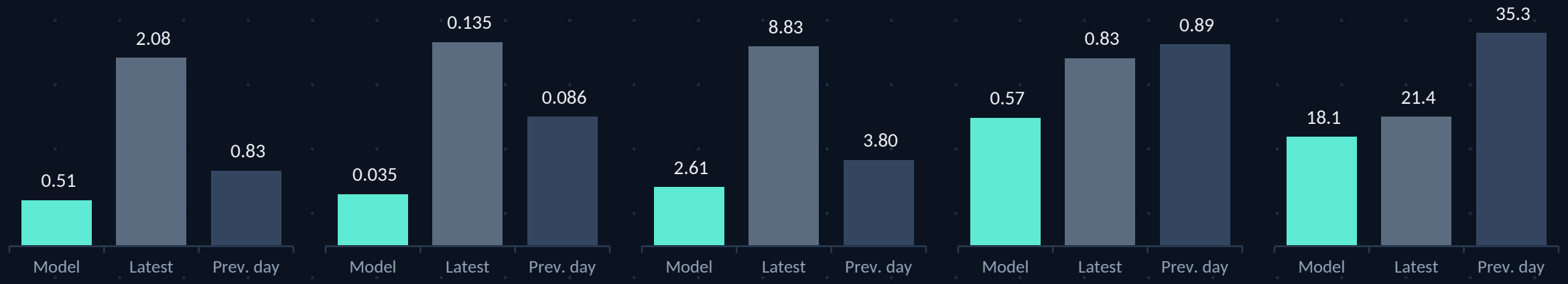
−32%

vs best baseline

Direction
MAE · degrees

−16%

vs best baseline



2025 test set · 8,757 scalar targets · direction: 8,593 non-calm targets (≥ 0.5 m/s), ideal decoding of quantized components

The design fits in a fifth of the Cyclone II

6,253

logic elements

19% of EP2C35F672C6

6

18 × 18 multipliers

12 of 70 nine-bit elements

50 MHz

physical test clock

+7.459 ns setup slack

79.74 MHz

fitted Fmax estimate

not a measured clock sweep

Flow Summary	
Flow Status	Successful - Tue Oct 27 20:50:14 2026
Quartus II 32-bit Version	13.0.1 Build 232 06/12/2013 SP 1 SJ Web Edition
Revision Name	weather_jtag
Top-level Entity Name	de2_weather_jtag
Family	Cyclone II
Device	EP2C35F672C6
Timing Models	Final
Total logic elements	6,253 / 33,216 (19 %)
Total combinational functions	4,147 / 33,216 (12 %)
Dedicated logic registers	4,571 / 33,216 (14 %)
Total registers	4571
Total pins	2 / 475 (< 1 %)
Total virtual pins	0
Total memory bits	0 / 483,840 (0 %)
Embedded Multiplier 9-bit elements	12 / 70 (17 %)
Total PLLs	0 / 4 (0 %)

Native Quartus II 13.0 Flow Summary · uninstrumented JTAG build · no values altered

Fit includes the board JTAG wrapper and excludes SignalTap. Timing panels in backup B5.

A forecast captured on the physical board



Native SignalTap capture, separately instrumented build · STP and decoded CSV retained

104

forecasts checked
from 128 input frames

temperature_out_q8

`01A72h` → `6770 / 256` → **`26.4453 °C`**

41 acquisition intervals ≈ documented 40-cycle RTL latency (sampling convention)

What we demonstrated, and what we did not claim

Demonstrated

- ✓ Five quantities forecast on the FPGA
- ✓ Recorded observations replayed over JTAG
- ✓ Fixed, pre-trained coefficients
- ✓ Physical replay at 50 MHz

Further work

- ○ Live sensor acquisition
- ○ Other sites and years
- ○ Rainfall probability
- ○ Power measurement and clock sweep

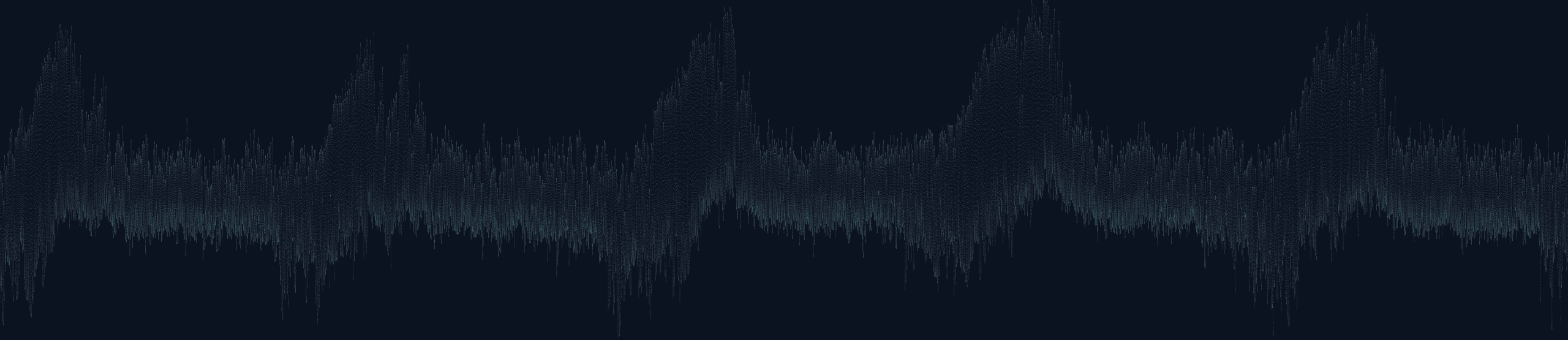
Hourly forecasting does not need sub-microsecond speed: the contribution is a verified hardware implementation and its schedule.

Weather inference, verified on FPGA

5
forecast quantities

0.80 µs
RTL latency at 50 MHz

104
physical forecasts checked



The original temperature model: a seasonal anchor and two corrections

$$\hat{T}(t+3) = T(t-21) + a \cdot \Delta\text{daily} + b \cdot \Delta\text{recent} + c$$

$$\Delta\text{daily} = T(t) - T(t-24)$$
$$\Delta\text{recent} = T(t) - T(t-3)$$

Parameter	Fitted value
a	0.7454870439
b	0.0091494450
c	0.0011295817 °C

Temperature test MAE **0.538017 °C** prototype → **0.507906 °C** current model

The temperature lane: seven deployed weights and a Q8 bias

Lag (h)	Input Q8	Weight Q14	Product
0	7680	20035	153868800
1	7552	−9335	−70497920
3	7168	1621	11619328
6	6912	−205	−1416960
12	6656	15	99840
21	7936	13908	110373888
24	7424	−10382	−77075968

SUM INCLUDING ALIGNED BIAS

$$S = 132050048$$

SCALE BACK TO Q8

$$\lfloor S / 16384 \rfloor = 8059$$

$8059 / 256 = 31.48\ ^\circ C$

Test errors keep the physical unit of each quantity

Target	MAE	RMSE
Temperature (°C)	0.507906	0.730939
Pressure (kPa)	0.034600	0.044638
Humidity (percentage points)	2.614292	3.685767
Wind speed (m/s)	0.566878	0.750218
Direction (degrees)	18.084598	30.201056

8,757 scalar targets. Direction: 8,593 targets with wind speed ≥ 0.5 m/s, ideal atan2 of quantized components. CORDIC maximum conversion error: 0.005635°.

Each lane repeats operand, product and accumulate for seven terms

Edge	Operation
`N`	Accept frame, shift history, initialize bias
`N + 1`	Capture lag-0 operand
`N + 2`	Capture product
`N + 3`	Add product to accumulator
`N + 4 … N + 21`	Repeat for the remaining six terms
`N + 22`	Scale, limit and publish lane result
`N + 23`	Earliest next eligible acceptance

Synchronous active-low reset cancels pending work and restarts history eligibility.

Native TimeQuest panels record the fitted timing margins

Multicorner Timing Analysis Summary

	Clock	Setup	Hold	Recovery	Removal	Minimum Pulse Width
1	Worst-case Slack	7.459	0.215	N/A	N/A	7.981
1	CLOCK_50	7.459	0.215	N/A	N/A	7.981
2	altera_reserved_tck	N/A	N/A	N/A	N/A	97.778
2	Design-wide TNS	0.0	0.0	0.0	0.0	0.0
1	CLOCK_50	0.000	0.000	N/A	N/A	0.000
2	altera_reserved_tck	N/A	N/A	N/A	N/A	0.000

Slow Model Fmax Summary

	Fmax	Restricted Fmax	Clock Name	Note
1	79.74 MHz	79.74 MHz	CLOCK_50	

PHYSICAL TEST CLOCK

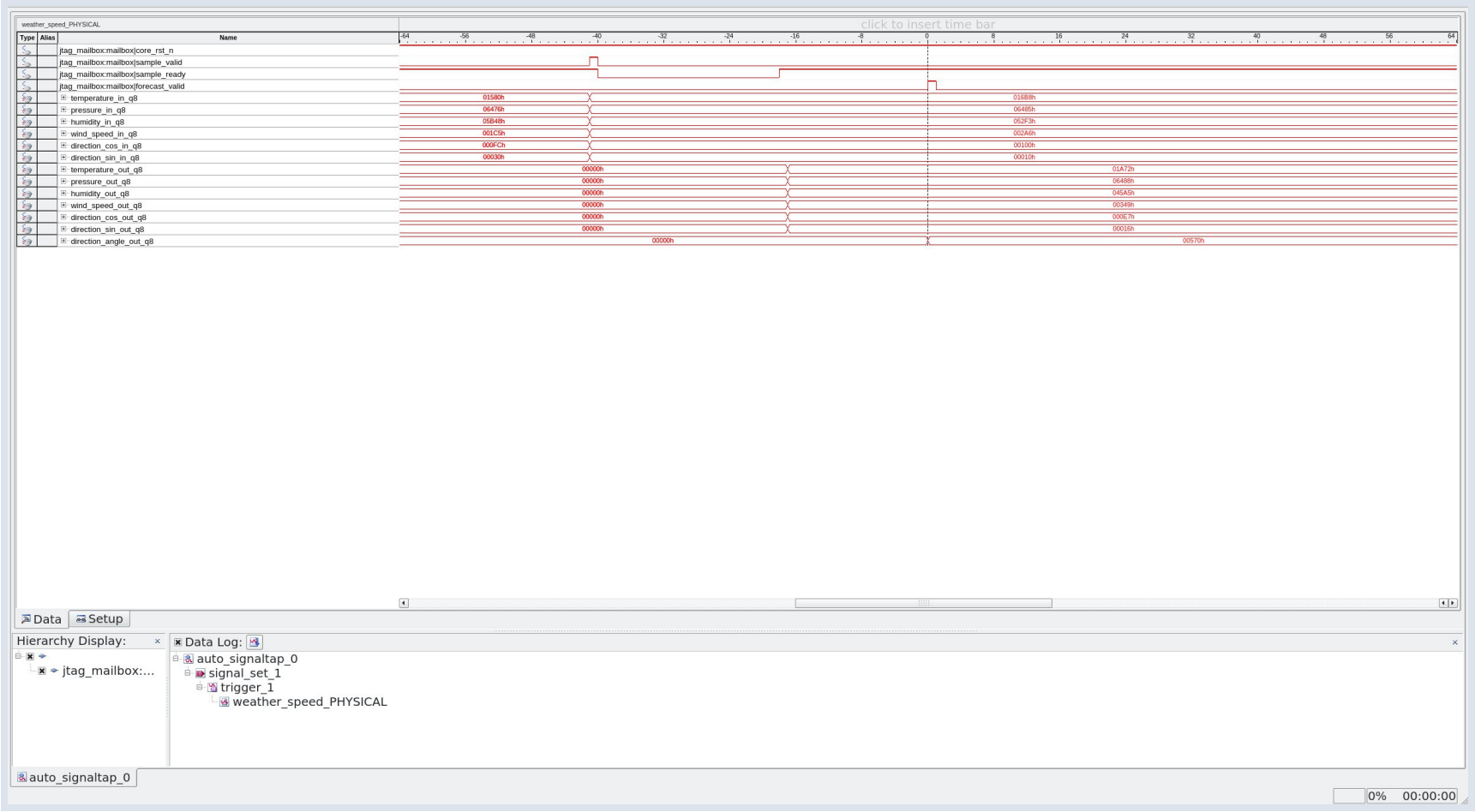
50 MHz

FITTED ESTIMATE

79.74 MHz

Setup slack 7.459 ns · hold 0.215 ns

The original SignalTap acquisition with decoded buses and trigger



The JTAG mailbox holds each request and keeps the completed forecast

01 **Stage payload** → 02 **Toggle request** → 03 **Wait for acknowledgement** → 04 **Read stable result**

128-bit
source word (host → FPGA)

256-bit
probe word (FPGA → host)

1
outstanding request at a time

Warm-up replies acknowledge without a forecast.

Source records behind the model, implementation and physical results

Model and evaluation

versions/weather_v2/models/model.json
versions/weather_v2/reports/metrics.csv
versions/weather_v2/data/weather.csv

Implementation

docs/reports/JTAG_IMPLEMENTATION_REPORT.md
docs/reports/weather_speed_modelsim_timing.json

Physical session

docs/reports/defense_session/
Native screenshots, replay logs, STP and decoded CSV